

An Analysis of Independent component analysis

Independent component analysis

Independent component analysis -- Part 1

Projection pursuit Signal mixtures tend to have Gaussian probability density functions, and source signals tend to have non-Gaussian probability density functions. Each source signal can be extracted from a set of signal mixtures by taking the inner product of a weight vector and those signal mixtures where this inner product provides an orthogonal projection of the signal mixtures. The remaining challenge is finding such a weight vector. One type of method for doing so is projection pursuit. Projection pursuit seeks one projection at a time such that the extracted signal is as non-Gaussian as possible. This contrasts with ICA, which typically extracts M signals simultaneously from M signal mixtures, which requires estimating a $M \times M$ unmixing matrix. One practical advantage of projection pursuit over ICA is that fewer than M signals can be extracted if required, where each source signal is extracted from M signal mixtures using an M -element weight vector. We can use kurtosis to recover the multiple source signal by finding the correct weight vectors with the use of projection pursuit. The kurtosis of the probability density function of a signal, for a finite sample, is computed as

Independent component analysis -- Part 2

General Derivation In the classical ICA model, it is assumed that the observed data $\mathbf{x}_i \in \mathbb{R}^m$ at time t_i is generated from source signals $\mathbf{s}_i \in \mathbb{R}^m$ via a linear transformation $\mathbf{x}_i = \mathbf{A} \mathbf{s}_i$, where $\mathbf{A}$ is an unknown, invertible mixing matrix. To recover the source signals, the data is first centered (zero mean), and then whitened so that the transformed data has unit covariance. This whitening reduces the problem from estimating a general matrix $\mathbf{A}$ to estimating an orthogonal matrix $\mathbf{V}$, significantly simplifying the search for independent components. If the covariance matrix of the centered data is $\Sigma_{\mathbf{x}} = \mathbf{A} \mathbf{A}^T$, then using the eigen-decomposition $\Sigma_{\mathbf{x}} = \mathbf{Q} \mathbf{D} \mathbf{Q}^T$, the whitening transformation can be taken as $\mathbf{D}^{-1/2} \mathbf{Q}^T$. This step ensures that the recovered sources are uncorrelated and of unit variance, leaving only the task of rotating the whitened data to maximize statistical independence. This general derivation underlies many ICA algorithms and is foundational in understanding the ICA model.

Independent component analysis -- Part 3

weighted by the mixing weights $a_{i,k}$. The same generative model can be written in vector form as $\mathbf{x} = \sum_{k=1}^n \mathbf{s}_k \mathbf{a}_k$, where the observed random vector $\mathbf{x}$ is represented by the basis vectors $\mathbf{a}_k = (a_{1,k}, \dots, a_{m,k})^T$. The basis vectors $\mathbf{a}_k$ form the columns of the mixing matrix $\mathbf{A} = (\mathbf{a}_1, \dots, \mathbf{a}_n)$. The basis vectors $\mathbf{a}_k$ form the columns of the mixing matrix $\mathbf{A} = (\mathbf{a}_1, \dots, \mathbf{a}_n)$ and the generative formula can be written as $\mathbf{x} = \mathbf{A} \mathbf{s}$, where $\mathbf{s} = (s_1, \dots, s_n)^T$. Given the model and realizations (samples) $\mathbf{x}_1, \dots, \mathbf{x}_N$ of the random vector $\mathbf{x}$, the task is to estimate both the mixing matrix $\mathbf{A}$ and the sources $\mathbf{s}$. This is done by adaptively calculating the $\mathbf{w}$ vectors and setting up a cost function which either maximizes the non-gaussianity of the calculated $s_k = \mathbf{w}^T \mathbf{x}_k$ or minimizes the mutual information. In some cases, a priori knowledge of the probability distributions of the sources can be used in the cost function. The original sources $\mathbf{s}$ can be recovered by multiplying the observed signals $\mathbf{x}$ with the inverse of the mixing matrix $\mathbf{W} = \mathbf{A}^{-1}$, also known as the unmixing matrix. Here it is assumed that the mixing matrix is square ($n = m$). If the number of basis vectors is greater than the dimensionality of the observed vectors, $n > m$, the task is overcomplete but is still solvable with the pseudo inverse.

Independent component analysis -- Part 4

Rescaling each vector $\mathbf{U}_i = \mathbf{U}_i / \sqrt{\mathbf{U}_i^T \mathbf{U}_i}$, and let $\mathbf{z} = \mathbf{U}$. The signal extracted by a weighted vector $\mathbf{w}$ is $y = \mathbf{w}^T \mathbf{z}$. If the weight vector $\mathbf{w}$ has unit length, then the variance of y is also 1, that is $\mathbf{E}[(\mathbf{w}^T \mathbf{z})^2] = 1$. The kurtosis can thus be written as:

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Independent component analysis -- Part 5

$$x_i = \bigvee_{j=1}^n (g_{ij} \wedge y_j), i = 1, 2, \dots, m, \quad \{\displaystyle x_{\{i\}}=\bigvee_{\{j=1\}}^{\{n\}}(g_{\{ij\}}\wedge y_{\{j\}}),i=1,2,\ldots,m,\}$$

Independent component analysis -- Part 6

$$H(\mathbf{Y}) = -\frac{1}{N} \sum_{t=1}^N \ln \left[\frac{p_{\mathbf{Y}}(\mathbf{y})}{p_{\mathbf{S}}(\mathbf{y})} \right] \quad \{\displaystyle H(\mathbf{Y})=-\frac{1}{N}\sum_{t=1}^N\ln\left\{\frac{p_{\mathbf{Y}}(\mathbf{y})}{p_{\mathbf{S}}(\mathbf{y})}\right\}\}$$

Independent component analysis -- Part 7

to achieve the independence of the extracted signal. If there are M marginal pdfs of the model joint pdf $p_{\mathbf{S}}$ are independent and use the commonly super-gaussian model pdf for the source signals $p_{\mathbf{S}} = \prod_{s=1}^M (1 - \tanh^2(s))^{-2}$, then we have